

Collaboration: The primary purpose of open source software is to encourage other developers to collaborate and contribute to the development of the software. This makes open source development, by nature, a community experience.

Modifiability: Open source software can be modified and re-distributed to solve different problems. This is the case with Linux, as its modified ‘distributions’ are used in various different gadgets and computers.

Decentralisation: One can argue that open source software is not inherently decentralised as making changes to the main project still requires approval of the project moderator(s) and owner(s). But if perceived from a different angle, given that the source code can be modified and re-distributed under proper licensing, this gives the developer the autonomy to use the software as and how desired thus making the process decentralised.

Convergent evolution

Blockchain and open source software are two distinct technologies with different themes and techniques for solving different challenges. However, when we compare the qualities of blockchain with open source, we begin to see parallels.

The expression ‘convergent evolution’ is used in evolutionary biology to describe the phenomenon in which different organisms from various time periods or epochs gain identical features. This occurs at the molecular level, as well as in animal morphology and even plant morphology. This is equally true of technological advancements. Every day, we create new technologies to meet a variety of issues; nevertheless, there are hidden patterns in the motivations that drive the development of these technologies.

Open source technology has been around for a lot longer than

blockchain. The emergence of open source technology, in my opinion, has had a direct impact on the birth of blockchain. Both technologies are centred on transferring power from the hands of a few to the hands of a large number of people. Both aim to improve product security and transparency, as well as encourage fairness and collaboration. While open source and blockchain have a wide range of uses, the motivations for their inception are similar. I believe it is safe to conclude that open source and blockchain share many qualities in common and are hence the result of convergent evolution in the world of technology.

How open source is driving the blockchain revolution

In this article, I hypothesised that blockchain and open source technologies are a product of convergent evolution. They share similar characteristics. But is open source also being used in the development of blockchain related technologies? The answer is a resounding YES.

- 1) Almost all the prominent cryptocurrencies in the market, including Bitcoin and Ethereum, are open source. Their design and codebases are available in the public domain.
- 2) ‘Hyperledger’, an initiative of the Linux Foundation, is leading the race in enabling the development of enterprise level private and permissioned blockchains for various different industries.
- 3) ‘Status’ is an open source project that uses Ethereum’s codebase to create a secure messaging and

eWallet app built on blockchain.

- 4) ‘Storj’ uses blockchain to build a secure and decentralised cloud storage system.
- 5) ‘High Fidelity’ employs blockchain to implement a virtual reality platform for hosting virtual events.
- 6) ‘Cosmos’ connects various services in different applications by implementing a blockchain to create a decentralised form of Internet.
- 7) ‘Aragon’ is built on Ethereum and uses blockchain to help create decentralised organisations.
- 8) ‘Original Protocol’ utilises Ethereum and IPFS to create decentralised online marketplaces using blockchain.
- 9) ‘Trust Wallet’ is a multi-platform cryptocurrency wallet built on blockchain.
- 10) ‘Augur’ is a market prediction platform developed on blockchain for placing wagers on market futures.

The projects listed above are some of the open source blockchain projects that are driving the most developer activity. These are some of the hundreds if not thousands of ways that blockchain is being implemented with the help of open source collaboration amongst developers across the world.

There’s no doubting that open source is a major driving force behind the rapid development and deployment of blockchains across a variety of industries. The creation of blockchains on such a vast scale would have been impossible without open source management systems like GitHub and BitBucket, as well as a version control system like Git. The blockchain revolution is being powered by open source. **END** 🐧

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